Sitting at my desk in the lab at 8 p.m., I typed *less Tsirt-JLS224.fastq* into terminal, and the ATCGs popped up on my computer screen for the first time. This was a garter snake transcriptome I had been working on for over a year. I had caught, processed, extracted RNA, library prepped, and sequenced this snake and 17 others. The excitement of discovery I felt in this moment is one reason I am pursuing a scientific career today. Seeing my data for the first time was a validating moment that I relive in my head whenever I feel frustrated with my progress or another line of code refuses to work. My motivation pushes me to overcome challenges and contribute to the scientific community through research, teaching, and outreach.

Many stigmas still surrounded women in STEM. After being at the University of South Alabama (USA) for 3.5 years, I have only taken one STEM class with a faculty member who identifies as a woman. Even more frustrating is that male faculty members in other STEM departments outside of biology have commented on my sex and how I am a weaker student and scientist because I am a woman. These persistent stigmas that "women do not belong in science" have strengthened my determination to continue this career path and support future women in STEM. The female faculty I have been able to work with have made my academic experience more fulfilling. Sadly, not all women in science get to interact with female STEM leaders. To combat these stigmas, I will continue to share the support I received during my undergraduate in my future Ph.D. program and career with all students and people I work with. My advocacy among the Biology Department, Honors College, and local community has contributed to women and minorities becoming more involved in STEM. My most impactful experiences have been working with K-12 students. The mother of 8-year-old Erin emailed me after I talked with her about snakes and turtles and said, "Our daughter was so excited and couldn't stop talking about how cool it was for a 'big girl' to love science as much as she does (Fig. 1)." Everyone deserves a role model, and to feel welcome in the STEM community. I will be part of the generation that prioritizes supporting anyone excited about learning and being involved in research.



Figure 1: Erin holding a garter snake for the first time after talking to me about her backyard biodiversity survey. Photo used with parent permission.

INTELLECTUAL MERITS

Differential venom gene expression in the Common Garter Snakes (*Thamnophis sirtalis*): My three years of undergraduate research in the Snakes, Spiders, Scorpions, INformatics, and Genomics (SSSTING) Laboratory working with Dr. Jason Strickland has been the most rewarding experience I have had as an undergraduate and solidified my desire to be a leader in research. When I first met with Dr. Strickland, he described three different projects, and I chose the one that would need to be built from the ground up, starting with specimen collection all the way to bioinformatic analysis. My Honors Thesis allowed me to conduct fieldwork, lab work, bioinformatics, and network with peers and faculty. Contributing to the first transcriptomic analysis of the genus and eventually seeing the culmination of my efforts being published are strong motivators. During the summer of 2021, I spent over 100 hours in the field to catch my target species. The bycatch records of mammals, reptiles, and amphibians I found in the field contributed to a 13-year longitudinal biodiversity study on my campus. Through this fieldwork, I earned co-authorship on a publication in September 2023 in the journal Herpetological Review (Canterbury et al., 2023).

My second year of research was spent milking snakes, dissecting specimens, and learning the molecular techniques needed to generate next-generation sequencing data. Lab work was followed by the part that has been the most exciting, bioinformatics and coding. DNA is life's blueprint, and bioinformatics allows me to conceptualize the blueprint and determine how changes in the genomic architecture result in variation. By the end of fall 2022, all samples were sequenced on an Illumina NovaSeq6000 and I was able to see my hard work pay off when I began processing my transcriptomic data. I have determined that three-finger toxins (3FTxs), snake venom metalloproteinases (SVMPs), and cysteine-rich secretory proteins (CRISPs) make up the majority of garter snake venom. The presence of 3FTxs, which are neurotoxins, is unique for the genus and suggests that *T. sirtalis*' diet has led to a unique venom profile. My research has

enabled me to network with many people and I have presented posters at two symposia for undergraduates and given oral presentations at two conferences: Alabama Chapter of Partners in Amphibian and Reptile Conservation and Southeast Partners in Amphibian and Reptile Conservation 2021. My involvement in undergraduate research has provided many opportunities to grow as a researcher and person.

NSF-REU at Cornell University and the Boyce Thompson Institute, "Identification of structural variants affecting fruit quality traits between wild and cultivated watermelons": Being one of the only students at the University of South Alabama working on bioinformatics drove me to apply for an REU at Cornell University. I was so excited to meet other students with similar interests and learn from some of the leading bioinformaticians in the country. I worked under Dr. Zhangjun Fei at the Boyce Thompson Institute to identify structural variants (SVs) in wild and cultivated Watermelon crops. Because of the challenges associated with polyploid genomes I needed to learn how to write my own Python and R scripts. I successfully identified 111,738 SVs which affected 19,482 and 19,096 genes in the cultivated and wild genomes, respectively. By integrating SVs and differential gene expression information, we identified candidate disease-resistant genes. The genome alignments I generated are being used in a pan-genome paper to continue the work I started on disease-resistance genes impacted by SVs. The SVs I identified also serve as a resource to inform watermelon breeding efforts.

While the REU project was intellectually challenging and invigorating, I faced challenges that motivated me to become a more independent scientist. Unfortunately, my PI had to spend the majority of the REU out of the country and we had little communication. My assigned Ph.D. student, Honghe, had never had a mentee before, and we struggled to communicate and stay on the same page. I was required to work more independently than my cohort at Cornell and learn how to problem-solve and reach out to other scientists. My peers were always surprised with how much solo work I was completing for my project. After finishing the project's original objective in two weeks, I pushed to expand the original idea allowing me to take more ownership of my project than I otherwise would have. Sharing my results and seeing my cohorts' projects at the symposium was the highlight of my REU. I am very proud of the work I was able to put in and the results that I generated. Although there were challenges, I became a better researcher and communicator because of them.

Tick salivary gland expression analysis: In collaboration with Dr. Strickland and Dr. Meghan Hermance in the College of Medicine at USA, I started as a paid research assistant in August of 2022 working on a project titled "A Ticking Time Bomb: Testing how viral infection modifies salivary gland gene expression in an invasive tick during feeding". Ticks are a major disease vector and are capable of spreading more pathogen species and greater pathogen diversity than any other arthropod group. I have been identifying salivary proteins that may facilitate virus transmission in the invasive Asian Longhorned Tick infected with Deer-Tick Virus. These gene candidates will be tested using a mouse model to determine if they facilitate viral transmission. I have been able to implement many of the gene-identifying scripts I created during my REU while still expanding my bioinformatic skills. I have gained experience working with new reference genomes and used new pipelines for testing differential expression compared to the ones I used working on the garter snake and watermelon projects. My ability to apply learned skills to new projects showcases the growth I have had as an undergraduate researcher.

BROADER IMPACTS

Mentorship: During my three years working in the SSSTING Lab, I mentored and advised seven incoming and current undergraduate students. Five of the seven incoming undergraduates are women, all with a different career direction. My mentorship included training them on literature review, RNA extraction and cDNA library prep, bioinformatic pipelines, and tutoring them for classes. I am so proud to see these students succeed and continue to share their experiences with others. The success I have seen from students in the lab encouraged me to mentor more incoming freshman, organize undergraduate research panels, and curate online resources for undergraduates who want to learn bioinformatics. In an attempt to expand outside of my University, I have curated a GitHub website as a resource for undergraduate students who

want to begin undergraduate research. I have included topics such as: writing an introductory email, getting started in bioinformatics, and picking a mentor. My fellow REU students began using my bioinformatic repository when I was at Cornell.

The success of my efforts encourages me to continue assisting undergraduates in their research journey, and my experiences working with underclassmen made me realize we need to engage students earlier in K-12. My experience teaching middle school classes confirmed this because, for many students, it was their first time feeling connected to their environment and the science community. My realization has led me to design a YouTube channel, "Learn about Biology through Bioinformatics," aimed at K-12 students to discuss new and cool research being published, how research is conducted, and how the students can become more involved in research and science.

Public outreach and teaching: My experience in community engagement has shaped the researcher and person I am today. Creating and managing our lab social media account has created a direct connection with the university and local community. The public has used our lab's Instagram account to ask questions about native species, venomous snake identification, and school science projects. I have used social media to intentionally engage broader communities by highlighting student and faculty researchers, events hosted by the biology department, and advertising national and local internships. I am particularly fond of our "women in stem" post that discusses real-world problems and highlights the women scientists in our lab and the research they are conducting (Fig. 2).



Figure 2: Social media post for International Women in Science Day where I highlighted all the women in the SSSTING Lab.

Meeting the people who interact with our social media at public events has been one of my favorite parts of public outreach. I have connected with our audience by speaking at community engagements and teaching local middle school classes. From teaching someone at the age of 65 about how their backyard wildlife is being affected by climate change to a child the age of 8 creating a "backyard biodiversity survey," I have gotten to share my love for science and learning how the world works with a variety of people. Seeing both older and younger generations connect with the local biodiversity is a privilege I am lucky to have. I am always excited to see who I will meet next at outreach events, and will continue engagment during my Ph.D. and career. I am also excited to expand my teaching experience to my fellow undergraduates. In the spring, I will be a Learning Assistant for the Advanced Bioinformatics course being taught by Dr. Strickland. As a learning assistant, I will be able to practice giving feedback on student code, host weekly help sessions, and give direction and advice for future coding experiences.

CAREER GOALS

I will earn a Ph.D. in Evolutionary Biology or a related field with an emphasis on bioinformatics working on molecular genomics and coevolution. After earning my Ph.D., I plan to work at a biotechnology company, such as Illumina, and develop novel methods to analyze genomic data including whole genome sequencing data and tissue-specific transcriptomes. During my career, I will expand the range of questions that researchers in both industry and academia can answer and further our ability to understand molecular genetics. With only 18.8 percent of Bioinformaticians being women, I will personally contribute to the increase of that number by joining the field and then recruiting other underrepresented groups. Additionally, I will continue my advocacy for women and minority involvement in STEM broadly.

Reference:

Canterbury, D.N., **G.E. Coppinger**, S. Morris, and J.L. Strickland. *Hyla gratiosa* (Barking Treefrog) Axanthism. *Herpetological Review* 54(2): 266-267.